

Amer Goel

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Education

University of Southern California

Los Angeles, CA

Doctor of Philosophy Candidate

- Department and School: *Data Sciences and Operations, Marshall School of Business*
- Group: *Operations Management*

Summer 2030

University of Michigan

Ann Arbor, MI

Bachelor of Science

- Major: *Mathematics: Mathematical Sciences, Sub plan Probabilistic Methods*
- Major: *Honors Linguistics*

May 2025

May 2025

GPA: 3.95/4.00

Relevant Coursework: Honors Multivariable/Vector Calculus, Proof-Based Linear Algebra, Abstract Algebra, Theoretical Statistics, Probability Theory, Real Analysis, Computational Linguistics, Logic, Syntax, Semantics, Phonology

Research Experience

NSF - REU Grant DMS2306672

Cambridge, MA

REU Researcher

Summer 2024

- Advised by **Dr. Aida Maraj** and **Alvaro Ribot** at Harvard University, proved combinatorial results in algebraic statistics related to convex polyhedra describing the feasibility of certain statistical estimators of statistical networks.
- Simulated graph structures, high-dimensional polyhedra, and path structures and dependencies arising in Gaussian Graphical Models and Colored Gaussian Graphical Models using SageMath and Python.
- Main Result: Inductive proof to characterize the closed form of halfspaces of shortest-path-polytopes over tree-graphs. The need for this closed form result comes from the inefficiency of available algorithmic methods, like Fourier-Motzkin or Double Description to recover the halfspaces of these polytopes.

Ongoing Projects

- Project advised by **Andrew Daw**. Started July 2025. Funded by NSF Grant 2441387.
 - Topics: Stochastic point processes, convex polyhedra, time-transformation, service operations.
- Project advised by **Yeganeh Alimohammadi**. Started August 2025.
 - Topics: Random graph optimization, bank networks, network contagion, network dynamics.

Papers

- ❖ Goel, A., Maraj, A., & Ribot, Á. (2024). *Halfspace Representations of Shortest Path Polytopes on Trees*. ms, Cambridge, MA. <https://arxiv.org/abs/2502.21204>
- ❖ Goel, A. (2025). *Simple and Periphrastic Past Tense Syntax in Catalan* (thesis). (N. Abner, Ed.).

Conferences

- ❖ Goel, A., Maraj, A., & Ribot, Á. (2025, January). Halfspace Representations of Shortest Path Polytopes on Trees. Joint Mathematics Meetings 2025. Seattle; WA.
- ❖ Goel, A. (2024g, November). *Algorithmic and Closed-Form Descriptions of Shortest-Path-Polytopes*. Gulf Coast Undergraduate Research Symposium. Rice University, Houston; Texas.

Presentations

- ❖ Goel, A. (2024, April). Fuzzy Sets: A Failed Cartesian Product. University of Michigan Directed Reading Program. Ann Arbor; MI.
- ❖ Goel, A. (2024, July). Shortest Path Polytopes of Statistical Models. University of Michigan REU. Cambridge; MA.
- ❖ Goel, A. (2024, July). Shortest Path Polytopes, The Toric Fiber Product, and Halfspace Construction. Harvard REU. Cambridge; MA.

Professional Experience

The Blackstone Group

Quant Summer Analyst

New York City, NY

Summer 2023

- Performed several advanced statistical analyses on US equities and commodities markets, including a principal component analysis coupled with linear regressions to determine sources of return variance
- Researched the effect of algorithmic trading on the economics of the underlying commodity markets being traded – concluded that large-volume trading does not show statistically significant impact
- Implemented trend-following price momentum signals including random walks, integrated processes, significance signals, and others based on moving statistics calculated from time series

AlixPartners LLC

Risk Advisory Consulting Intern

Chicago, IL

Summer 2022

- Performed rigorous accounting analyses of international financial statements, including a 245-month discounted cash flow analysis, cost, revenue, and operations comparison analyses, and other comparison-based valuations to quantify economic damages for AlixPartners clients
- Analyzed 1000+ pages of Spanish-language international documents, including original arbitration contracts, equity contribution agreements, bidding agreements, and complex financial statements
- Managed, summarized, and visualized complex data for several disputes, including official and classified conflicts from a large energy cooperative and a multinational airport concessionaire

Skills

- *Computational*: Basic proficiency in Java, C++, proficiency in LaTeX, Python, SageMath, Excel, Powerpoint,
- *Mathematics*: Combinatorics, affine geometry, combinatorial geometry, graph theory
- *Language*: Fluent: Spanish; Highly Proficient: Portuguese, Catalan; Basic Proficiency: Italian, French, Hindi, Urdu

Undergraduate Leadership

Michigan Undergraduate Law Review

Editor-in-Chief

Ann Arbor, MI

January 2022-December 2024

- Used WestLaw, LexisNexis, Bloomberg Law, and academic journals in English, Spanish, and Portuguese to write and edit legal research pieces about statistical methodology in law and enforcing international treaty law
- Developed proprietary legal curriculum to educate writers and editors on academic standards, including Bluebook citation manual, legal theory, legal language and interpretation, and analytical court opinion readings
- Led journal of 50 staff members including a full-length law review, two teams of editors, and a blog product by reviewing legal pitches, editing content, managing editor relationships, and proofreading all published articles

The Michigan Daily

Research Reporter | Investigative Journalist

Ann Arbor, MI

December 2022-May 2025

- Investigated research and administrative topics in STEM, social science, and humanities by connecting with 30+ University of Michigan professors, researchers, academic writers, and student leaders
- Conducted official journalistic interviews with scientists, administrators, industry experts, and leaders of student organizations to report on cutting-edge academic and medical research conducted at the University of Michigan
- Investigated confidential university-related academic misconduct with a team of other investigative reporters using FOIA requests, trauma-informed reporting, and court documentation tools like PACER and MiCOURT

Michigan Journal of International Affairs

Head of Finance | Staff Writer

Ann Arbor, MI

September 2022-May 2024

- Researched current international affairs in Latin America, particularly from primary Spanish-language sources including local news, press releases, and official political correspondents
- Conducted independent investigations in econometrics, comparative law, and international policy to show how social factors like history, commerce, and language affect the current politics of the Western Hemisphere
- Directed the finances of the organization by budgeting for official club events, including social events, merchandising, and organizing logistics for club-sponsored bonding events

Michigan Journal of Economics

Vice President | Head of Research

Ann Arbor, MI

September 2021-May 2024

- Collaborated with peers to explore fringe topics in economics including behavioral economics, sports economics, and theory, using tools from mathematical modeling and game theory
- Lectured and developed a curriculum for economic research writing, teaching new writers how to interpret, interact with, and produce formal economic writing to prepare them for a complete supervised research project
- Led several workshops in professional development, including LinkedIn format, Resume and CV making, network development, and professional communication for members recruiting private-sector jobs

Undergraduate Journalism

❖ Research Journalism

- ❖ Goel, A. (2022, March 12). An Exploration in Mathematical Cartography: Comparative Distortion. Ann Arbor. *DeepBlue Repository*.
- ❖ Goel, A. (2023). A Mathematical Characterization of Some Judicial Philosophies. *Michigan Undergraduate Law Review*, 1(1), 63–76.
- ❖ Goel, A. (2023). An Algebraic Presentation of Market Structure. *Michigan Journal of Economics*.
- ❖ Goel, A. (2023). Bioeconomics: Cyclical Equilibrium in a Sequential-Game Ecosystem. *Michigan Journal of Economics*.

❖ Social Science

- ❖ Goel, A. (2022). In Defense of Rationality. *Michigan Journal of Economics*.
- ❖ Goel, A. (2022). Neuroeconomics: The Chemistry of Rationality. *Michigan Journal of Economics*.
- ❖ Goel, A. (2023). I'm Going to Steal the Moon: Evaluating the Protection of the Common Heritage of Mankind. *Michigan Undergraduate Law Review*, 1(2), 35–44.
- ❖ Goel, A. (2023). The Colombia/Venezuela Border Reopening: Was it Worth It? *Michigan Journal of International Affairs*.
- ❖ Goel, A. (2023). What You Think You Know (But Really Don't) About Tax Havens. *Michigan Journal of International Affairs*, 26, 18.
- ❖ Goel, A. (2024). Scripta Non Manent. *Michigan Journal of International Affairs*, (27), 15.
- ❖ Goel, A. (2024). It's Complicated. *Michigan Journal of Economics*.
- ❖ Goel, A. (2024b). Make Room, Dutch! Sranan Tongo as an Official Language of Suriname. *Michigan Journal of International Affairs*, (28), 21.

❖ News

- ❖ Goel, A., & Moncke, S. (2022, January 18). Ono Outlines Dei Goals, Announces Search For Sustainability Director In State Of The University Address. *The Michigan Daily*.
- ❖ Goel, A., & Moncke, S. (2022, November 7). Sacua Talks Availability Of Meeting Minutes, Faculty Childcare. *The Michigan Daily*.
- ❖ Goel, A. (2023, January 11). International Research Study Finds New Genetic Links To Heart Disease. *The Michigan Daily*.
- ❖ Goel, A. (2023, January 11). President Ono And Chief Diversity Officer Tabbye Chavous Discuss Dei Initiatives On Campus. *The Michigan Daily*.
- ❖ Goel, A. (2023, January 24). Infectious Diseases Covid-19, Rsv, Influenza Still Prevalent On Campus. *The Michigan Daily*.
- ❖ Goel, A. (2023, February 6). What Does Umich Do With Its Particle Accelerator? *The Michigan Daily*.
- ❖ Goel, A. (2023, March 22). Behind The B: Foundations Of Cryptocurrency. *The Michigan Daily*.
- ❖ Goel, A. (2023, March 29). The ABCs Of Umich Oncology: Activism, Bioinformatics, Collaboration. *The Michigan Daily*.
- ❖ Goel, A. (2023, April 9). Umich Perspectives: How Are We Dealing With AI? *The Michigan Daily*.
- ❖ Goel, A. (2023, April 30). GEO Continues Strike, Grading Plans Remain In Flux. *The Michigan Daily*.
- ❖ Goel, A., Spring, E., & Johnston, B. (2023, September 21). Umich Becomes First Major University To Develop Its Own Generative AI. *The Michigan Daily*.
- ❖ Goel, A. (2023, September 27). Umich Chosen To Lead \$30 Million Complex Particle Research Center. *The Michigan Daily*.
- ❖ Goel, A. (2023, October 1). September Lab Spotlight: Kamcev Research Laboratory. *The Michigan Daily*.
- ❖ Goel, A. (2023, March 8). One For The Books: How Umich Libraries Have Become More Than Just Places To Read. *The Michigan Daily*.

Awards and Interests

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| ❖ GCURS Outstanding Session Presentation, Computational Math and Operations Research | ❖ Cartography |
| ❖ University Highest Honors in Linguistics (based on merit of honors thesis) | ❖ Billiards |
| ❖ NSF Graduate Research Fellowship (Industrial Engineering and Operations Research) | ❖ Baking & Cooking |
| | ❖ Art & Architecture |
| | ❖ Poetry |